

Realistic Simulation of Galewsky Initial Value Problem

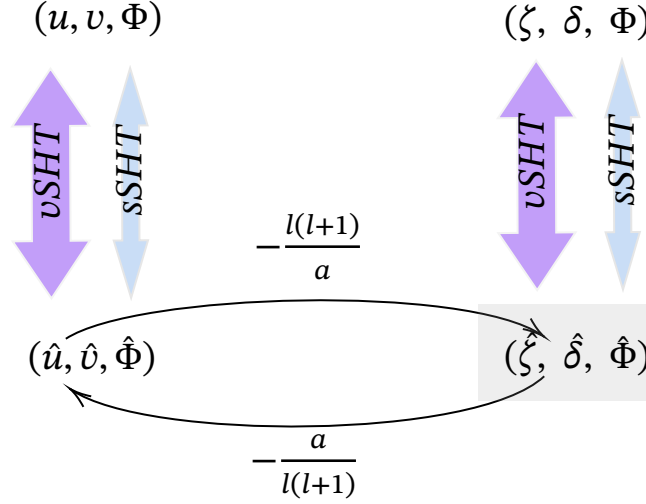
Haoyu Tang

July 4, 2026

Contents

1 Psuedospectral Method

There are four equivalent ways of describing the SWE system on the sphere:



The transformation between $(u, v, \Phi) \leftrightarrow (\hat{u}, \hat{v}, \hat{\Phi})$ and similarly for $(\zeta, \delta) \leftrightarrow (\hat{\zeta}, \hat{\delta})$ is the real spherical harmonic transform. In particular, for $(u, v) \leftrightarrow (\hat{u}, \hat{v})$, we use the **real vector spherical harmonic transform** defined as:

Def. 1. Let $\mathbf{v}$ be a vector field $S^2 \rightarrow \mathbb{R}^2$. Consider the multipole expansion:

$$\mathbf{v} = \sum_{l=1}^{\infty} \sum_{m=-l}^l \psi_{lm} \Psi_{lm} + \phi_{lm} \Phi_{lm} \quad (1)$$

where

$$\begin{cases} \Psi_{lm} &= \nabla Y_{lm}(\theta, \phi) \\ \Phi_{lm} &= \mathbf{k} \times \nabla Y_{lm}(\theta, \phi) \end{cases}$$

where Ψ_{lm} is the curl-free **Poloidal Harmonics**. Φ_{lm} is the divergence-free **Toroidal Harmonics**. $\mathbf{k}$ is the unit vector pointing outward from the surface. Define the vector spherical harmonics transform $\mathcal{F}_{vSHT}$ as:

$$\mathcal{F}_{vSHT} : L^2(TS^2) \rightarrow C(A; \mathbb{C}) \quad (2)$$

where TS^2 is the tangent bundle on S^2 and $A = \{(l, m) : l \in \mathbb{N}^+, -l \leq m \leq l\}$

Some useful properties of vector spherical harmonics:

Azimuthal symmetry up to sign and complex conjugation:

Prop. 1.

$$\Psi_{l,-m} = (-1)^m \Psi_{l,m}^* \quad (3)$$

$$\Phi_{l,-m} = (-1)^m \Phi_{l,m}^* \quad (4)$$

Orthogonality:

Prop. 2.

$$\langle \Psi_{lm}, \Psi_{l'm'} \rangle = l(l+1) \delta_{ll'} \delta_{mm'} \quad (5)$$

$$\langle \Phi_{lm}, \Phi_{l'm'} \rangle = l(l+1) \delta_{ll'} \delta_{mm'} \quad (6)$$

$$\langle \Psi_{lm}, \Phi_{l'm'} \rangle = 0 \quad (7)$$

The first property allows us to cleanly define the truncated version of vSHT:

Def. 2. Define the vector spherical harmonics transform as:

$$\text{vSHT}^{n_g, l_{\max}, m_{\max}} : n_g \times 2n_g \rightarrow l_{\max} \times m_{\max} \quad (8)$$

$$\{(u_i, v_j)\} \mapsto \{\phi_{lm}, \psi_{lm}\} \quad (9)$$

where n_g defines the grid size corresponding to latitude and $l_{\max}, m_{\max}$ specifies the truncation of the multipole expansion.

The second property gives an easy transformation between $(\hat{u}, \hat{v}) \leftrightarrow (\hat{\zeta}, \hat{\delta})$, recall:

Prop. 3.

$$(\hat{\zeta}, \hat{\delta}) = -\frac{l(l+1)}{a} \mathcal{F}_{\text{vSHT}}(u, v)$$

With the inverse transformation given by:

$$(\hat{u}, \hat{v}) = -\frac{a}{l(l+1)} \mathcal{F}_{\text{vSHT}}(\hat{\zeta}, \hat{\delta})$$

by applying the definition $\zeta = k \cdot \nabla \times \mathbf{v}$, $\delta = \nabla \cdot \mathbf{v}$ to the multipole expansion of $\mathbf{v}$